Mini Dino Game

Assembly, Wiring and Gameplay Guide

Arduino Uno + 16x2 I2C LCD + Rotary encoder + Buzzer

What you are building

A small Chrome-dino style game. The dino sits at the left of the LCD, a cactus scrolls in from the right, and you press the rotary encoder to jump over it. Survive and the score climbs; hit a cactus and the game ends. Everything is solderless - modules plug into the breadboard and the Uno, and jumper wires make the connections.

The dino and cactus are drawn with custom 5x8 characters held in the LCD's own memory. A 16x2 character LCD cannot show large graphics, so the game is deliberately small: two run frames for the dino, one cactus, and a score line.

Parts you have

Item	Qty	Price (Rs.)
Arduino Uno R3, with USB cable	1	280
16x2 LCD with I2C backpack, 4-pin	1	185
Rotary encoder with push switch	1	150
Breadboard, 400 point	1	50
Jumper wires, male-to-male	1	50
Jumper wires, male-to-female	1	50
Passive piezo buzzer	1	10
Total	**7**	**775**

These are the prices on your Chandni Chowk bill. Total Rs. 775 - inside the Rs. 600-800 budget.

How it connects

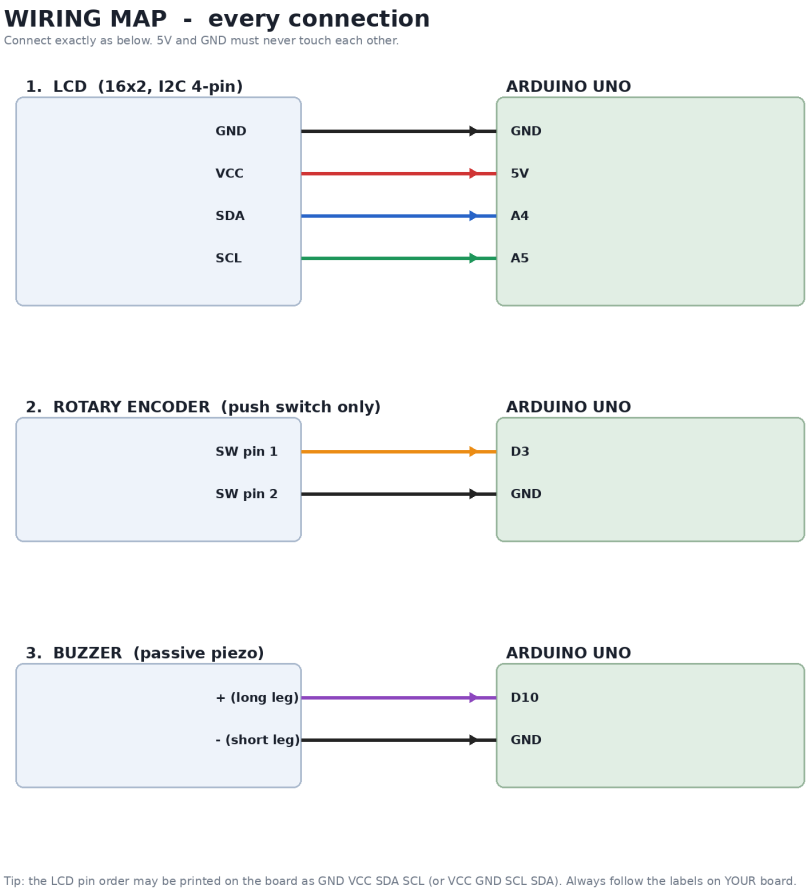
The LCD takes four wires on the I2C bus: GND, VCC, SDA to A4 and SCL to A5. The encoder contributes only its push switch - two pins, to D3 and GND. The buzzer takes two wires, + to D10 and - to GND. Eight wires in total, and no component needs a resistor, a capacitor or a soldering iron.

Before you start

Keep the board unplugged until every wire is in place. Trace each connection against the wiring map on the next page before you apply power, and make sure the 5V and GND wires never touch each other - a short across those two rails can damage the Uno.

Wiring: every connection

Make these eight connections. The wire colours are only a suggestion - they make mistakes easier to spot, but any jumper wire works. Check the labels printed on your own LCD, because the pin order on the 4-pin header differs between modules.



Wiring map. Each module connects to the Uno exactly as shown.

Module	Module pin	Arduino Uno pin	Wire colour
LCD	GND	GND	black
LCD	VCC	5V	red
LCD	SDA	A4	blue
LCD	SCL	A5	green
Encoder	Switch pin 1	D3	orange
Encoder	Switch pin 2	GND	black
Buzzer	+ (long leg)	D10	purple
Buzzer	- (short leg)	GND	black

Assembly, step by step

Work through the steps in order. Steps 1 to 7 are the hardware; do not plug in the USB cable until step 8.

1. Lay out the parts. Put the Uno and the breadboard side by side. Keep the Uno's USB socket facing you so the power and ground pins are easy to reach.

2. Read the LCD pin labels. Look at the 4-pin header on the LCD's I2C backpack. The board is printed with GND, VCC, SDA and SCL, but the order varies between modules - write yours down. Ignore the long 16-pin header on the other edge; it is not used.

3. Connect the LCD, four wires. Use male-to-female jumpers: the female end clips onto the LCD pins, the male end goes into the Uno's female headers. GND to GND, VCC to 5V, SDA to A4, SCL to A5.

4. Find the encoder's switch pins. The encoder has five pins. Three belong to the rotation (A, C, B) and two to the push switch. On most encoders the switch pins sit on the opposite side from the three rotation pins. If you have a multimeter, set it to continuity and press the shaft - the two pins that beep are the switch.

5. Connect the encoder, two wires. Run one switch pin to D3 and the other to GND. Leave the three rotation pins unconnected; this game uses only the push switch. The switch has no polarity, so either pin can go to either place.

6. Connect the buzzer, two wires. The long leg is positive: take it to D10. The short leg goes to GND. If your buzzer has a red and a black lead, red is positive.

7. Check before powering up. Trace every wire once against the wiring map. 5V and GND must never touch, and no bare wire end should bridge two pins on the Uno.

8. Plug in the USB cable. Connect the Uno to your computer. The board's power LED should light. If nothing lights, the cable is probably charge-only - swap it for a data cable.

Software: install and upload

1. Install the Arduino IDE from arduino.cc, then open it.
2. Go to Tools > Board and choose Arduino Uno.
3. Go to Tools > Port and choose the port that appeared when you plugged the board in.
4. Go to Tools > Manage Libraries, search for LiquidCrystal I2C (by Frank de Brabander) and click Install.
5. Open MiniDino.ino. The sketch folder must be named Dino and must contain MiniDino.ino.
6. Click Verify (the tick) and wait for Done compiling.
7. Click Upload (the arrow) and wait for Done uploading - the screen will start showing the game.

What the game shows on the screen

The LCD has two lines of 16 characters. The bottom line is the play field; the top line carries the score and the messages. Nothing else is drawn, which keeps the game readable on a small character display.



Start screen

Start screen - the game waits for a button press.



Playing - dino (left) + cactus scrolling, score top-right

Playing - the dino stands at the left, the cactus moves in from the right, the score sits top-right.



Game over

Game over - the final score, and a low beep from the buzzer.

How the dino and cactus are drawn

A 16x2 LCD stores up to eight custom characters, each 5 pixels wide and 8 tall. The sketch defines three of them: two dino run frames - the legs alternate, which is what makes the dino look like it is running - and one cactus with two arms. During play the cactus moves one character cell to the left every 230 milliseconds; the dino stays in column 0 and jumps to the top row for 450 milliseconds when you press the button.

The sketch redraws only the cells that changed - the old cactus position, the dino and the score - instead of clearing the whole screen. On a character LCD a full clear flickers badly, so this detail is what keeps the display steady.

How to play

1 Press the encoder once. The start screen changes to the running game.

2 Press again while the dino is running. The dino jumps to the top row and the buzzer beeps.

3 Let the cactus pass under the dino. Each cactus you clear adds one point.

4 If the cactus reaches column 0 while the dino is on the ground, the game ends.

5 Press once more after game over to return to the start screen.

Two numbers you can tune

Setting	In the sketch	Effect
Game speed	MOVE_MS = 230	Lower is faster. Try 180 for harder, 300 for easier.
Jump time	JUMP_MS = 450	Longer keeps the dino in the air longer.

If something does not work

Symptom	Likely cause	Fix
Screen completely blank	Contrast pot, or wrong I2C address	Turn the blue pot on the back of the LCD. If still blank, change 0x27 to 0x3F in the sketch and re-upload.
Only dark blocks on screen	Contrast set wrong	Turn the pot slowly in both directions until the text appears.
No game after upload	Wrong board or port	Check Tools > Board is Arduino Uno and the correct Port is selected, then upload again.
Button does nothing	Wrong pair of encoder pins	Move the two wires to the other two encoder pins. Only the push-switch pins work.
Buzzer is silent	Polarity reversed, or active buzzer	Swap the two buzzer wires. If it still gives one flat beep, it is an active buzzer - the sketch needs a passive one.
Upload fails	Charge-only cable or wrong port	Use a data cable and pick the port that appears when the board is plugged in.
Game restarts on its own	Loose jumper or weak USB power	Reseat every jumper, then power the board from a powered USB port.

Reference

Sketch: MiniDino.ino, about 170 lines. Library: LiquidCrystal I2C. Board: Arduino Uno R3. I2C pins on the Uno: SDA = A4, SCL = A5. The default address of these backpacks is 0x27, or 0x3F on some units.

If you later want the larger graphics version, the same board runs a 128x64 OLED in place of the 16x2 LCD, which allows the full dino and cactus sprites instead of 5x8 characters.